Total Printed Pages: 02

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EIA-101/ECA-101
Examination –Dec- 2022
B.Tech. I Sem : E&I, EC
Basic Electronics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a – Max 50 words, Part b – Max 50 words,
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

Q.1 (a) Give the applications of LASER DIODE. 02

(b) For tunning circuit which type of diode is used? Justify your answer. 02

(c) Compare Zener and avalanche breakdown. 03

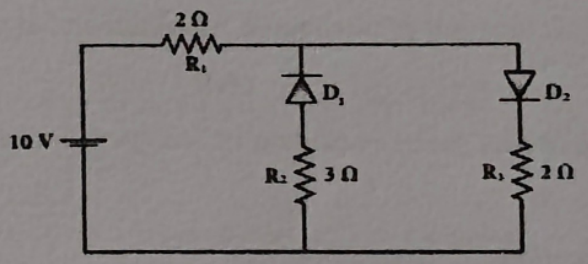
OR

Draw the VI characteristics of tunnel diode. 03

(d) With the help of VI Characteristics explain the working of Zener diode. 05

OR

The given circuit has two ideal diodes connected as shown in figure below. Calculate the current flowing through the resistance R1 . 05



Q.2 (a) Define punch through. 02

(b) Derive the relation between alpha and beta. 02

(c) Why common base transistor input characteristics moves inward give reasons. 03

OR

What causes collector current to flow when the emitter current is zero and what is this collector current called? 03

(d) Sketch the Circuit diagram of a NPN common base configuration explain its working and VI characteristics. 05

OR

With the help of circuit diagram explain any two transistor biasing circuits. 05

Q.3 (a) Give the advantage of h-parameter analysis. 02

(b) Define the four h-parameter. 02

(c) Draw the hybrid equivalent circuit for common base transistor. 03

OR

Draw the hybrid equivalent circuit for common emitter transistor. 03

(d) Derive the equations for current gain and voltage gain of a CE amplifier using h-parameters. 05

OR

Give low frequency h-parameter model of BJT and explain the same. 05

Q.4 (a) List the applications of Direct coupling. 02

(b) Differentiate between cascade and cascode amplifier. 02

(c) Derive the equation for the overall voltage gain of multistage amplifier in terms of the individual voltage gain. 03

OR

State Millers Theorem. 03

(d) Draw the circuit diagram of Darlington Amplifier and explain. 05

OR

With the help of neat sketches explain RC coupled amplifier. 05

Q.5 (a) List the applications of tuned amplifiers. 02

(b) Discuss crossover Distortion. 02

(c) Compare single tuned and double tuned amplifiers. 03

OR

Discuss harmonic Distortion. 03

(d) Discuss the characteristics and response of double tuned amplifiers. 05

OR

Differentiate A,B,AB and C type of power amplifier. 05

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